

1 Superposition Codes and Sparse Recovery

We begin by describing the toy scenario to be studied. Given a large number N , consider a map F that represents each subset $x \subseteq [N] = \{1, \dots, N\}$ by a linear combination

$$y(x) = \sum_{i \in x} f_i \in \mathbb{R}^d,$$

where the vectors $\{f_i \in \mathbb{R}^d : i \in [N]\}$ are chosen in advance and where the dimension d of the encoding is expected to be much smaller than N . The vectors f_i are called codewords for the elements of $[N]$, the collection $\{f_1, \dots, f_N\}$ is called a dictionary, and the image Fx is called a superposition code. It is useful to view x as a vector in $\{0, 1\}^N$ with coefficients

$$x_i = \begin{cases} 1 & : i \in x \\ 0 & : \text{otherwise} \end{cases}$$

and view F the matrix of column vectors $[f_1 \dots f_N]$, so that $y = Fx$ is a linear equation. In this work, we'll model our subset as a random variable X uniformly distributed over all subsets of some fixed size $k \ll N$. Keeping with the orders of magnitude discussed in Gao et al. (2024), we consider $N \approx 1000,000$ and $k \approx 100$.

The problem of recovering a sparse signal x from a linear projection Fx is known as sparse recovery. This problem is considered in the subject of compressive sensing. In this work, we will use language from coding theory and think of x as some data to be encoded and the projection $y = Fx$ as a code. Thus, sparse recovery becomes the task of “decoding” x . Note that this is opposite to the convention used in sparse autoencoders, where the map estimating x from y is called the “encoder.” (In classical applications of autoencoders, the learned representation is understood as an efficient “code” for the data distribution.)

Due to a connection with sparse linear regression, the subproblem of identifying the support of x from the projection Fx is often called model selection. When the coordinates of x take values in $\{0, 1\}$, sparse recovery reduces to a problem of model selection. In general, note that model selection presents the main difficulty of sparse recovery; once the support of x identified, its non-zero coordinates can be estimated easily e.g. by solving a least squares problem.

It remains to choose the dictionary F and to decide which algorithms we will use for sparse recovery. In practice, the dictionaries used by neural networks will likely have special properties related to the nature of the data being encoded and the operations that need to be performed. However, for the purposes of encoding a uniformly random subset of $[N]$, one natural way to choose a dictionary is to assign each codeword f_i to a random unit-norm vector. The simplest motivation for this strategy is the fact that, so long as $d = \Omega(\ln N)$, we can expect a dictionary of N randomly chosen vectors to be a “nearly orthonormal” frame in the following sense.

Proposition 1 *For $\epsilon > 0$ and $p > 0$, let $d > 2\epsilon^{-2}(2 \ln N + \ln p^{-1})$ and let*

$$\{F_1, \dots, F_N\} \subseteq \{-1/\sqrt{d}, 1/\sqrt{d}\}^d$$

be random vectors with independent, uniformly distributed entries. Then

$$\forall i \neq j, |\langle F_i, F_j \rangle| < \epsilon$$

with probability at least $(1 - p)$.

This justifies that, for fixed k , we need only take $N = O(\ln N)$ in order for each coefficient x_i to be retrievable directly by thresholding the linear function $\langle y, f_i \rangle$ of the code y . We informally refer to decoding methods that require only a single “step” of estimation and thresholding as one-step methods. On the other hand, the field of compressive sensing provides a number of more elaborate *iterative* methods for sparse recovery. Some of these are based on minimizing an objective function like

$$\|Fx - y\|_2^2 + \lambda \|x\|_1$$

for some $\lambda > 0$, which in statistics is known as lasso regression. However, many of these methods are not applicable to the very large problems faced by sparse autoencoders, where performing a training run using only a one-step method is already computationally demanding.

1.1 One-Step Methods

One way to motivate the construction of one-step methods is to think of the isolated subproblem of estimating a coefficient X_i from the model

$$Y = X_i f_i + \sum_{j \neq i} \overbrace{X_j f_j}^{Z_i}. \quad (1)$$

(Here we write X and Y with capital letters to emphasize they are random variables.) If we approximate the sum Z of other codewords as centered Gaussian noise with covariance Σ , the inference problem Y_i becomes tractable; specifically, if we define an inner product by $\langle v, w \rangle_\Sigma = \frac{1}{2} v^T \Sigma^{-1} w$, then the posterior of Y_i conditional on X is determined by the function

$$\lambda_i(Y) = \frac{\langle f_i, Y \rangle_\Sigma}{\|f\|_\Sigma^2}.$$

In signal processing, this kind of linear measurement is called a **matched filter** for the codeword f_i . The posterior on X_i is given in terms of the matched filter by

$$\ln P(Y_i = y | X = x) = \frac{\rho}{2} (\lambda_i(x) - y)^2 + \ln P(Y_i = y) + C \quad (2)$$

where $\rho = \|f\|_\Sigma^2$ is called the “signal-to-noise ratio” of the filter λ . See Appendix A for a review.

When f_i are independent draws from some isotropic distribution, it is natural to take Σ to be some multiple of the identity, so that our decoder has the form

$$\hat{X}_i(Y) = \begin{cases} 1 & : \langle f_i, Y \rangle \geq \tau \\ 0 & : \text{otherwise} \end{cases}$$

for some $\tau \geq (0, 1)$. If the number k of non-zero latents is known or can be guessed in advance, one simple improvement is to choose τ adaptively so that exactly k of the estimates $\hat{X}_i$ are non-zero. This is called top- k decoding, and was introduced in Makhzani & Frey (2014). Gao et al. (2024) showed that, in practice, top- k autoencoders perform better than their ReLU variants in practice.

1.2 Matching- k

The second class are the iterative methods.

We’re interested in understanding how large the dimension d needs to be as a function of (N, k) for a given inference method to recover Y . (We do not study the problem of learning the dictionary.) Since Y is a discrete variable, we will focus on conditions for *exact* recovery. We focus on a regime where Y resembles the very sparse latent representations learned by sparse autoencoders trained on large language models. For example, typical values discussed in Gao et al. (2024) are $N = 2^6$ and $k = 100$.

2 Information Theory Bounds

Suppose that the latent variable Y to be encoded is a symbol drawn from an alphabet of size N . What is the minimum dimension d such that each value of Y can be represented by a vector $X \in \mathbb{R}^d$?

When the vector X can be stored exactly, the answer to our question simply depends on the number of values it can take. If the coefficients of X are 16 bit floating point numbers, then each coefficient can store (nearly) 2^{16} distinct numbers, and overall we need only about $d_{\min} \approx \frac{1}{16} \log_2 N$ dimensions.

However, we expect that activation vectors within real neural networks are subject to various kinds of interference. For example, an activation vector X may need to represent the symbol Y while also holding information on another latent quantity, or it may be subject to explicitly induced noise like dropout. Characterizing the “capacity” of a vector in the presence of interference is a basic problem in coding theory.

One common model is to consider that X is subject to additive white Gaussian noise. More specifically, we let $Z \in \mathbb{R}^d$ be a vector of independent, centered Gaussians each of variance P , and consider the problem of recovering Y from $X + Z$. The following is then a typical (but remarkable) result of information theory.

Proposition 2 *Let the random variable $Y \in [N]$ be uniformly distributed, and let Z be white Gaussian noise of variance P . Suppose there exist a pair of maps $F: [N] \rightarrow \mathbb{R}^d$ and $G: \mathbb{R}^d \rightarrow [N]$ so that*

$$G(F(Y) + Z) = Y$$

with probability at least $(1 - p)$, and suppose the variance of each coordinate of $F(Y)$ is bounded by 1. Then

$$d \geq 2 \frac{(1 - p) \ln N - \ln 2}{\ln(1 + P^{-1})}.$$

In particular, if we fix P and let the number N of symbols grow to infinity, this shows that the minimum dimension d of a reliable code must grow logarithmically in N . More specifically, we can say the following.

$$\frac{d}{\ln N} \geq (2)$$

Since $\ln(1 + P^{-1})^{-1} = P + O(P^{-2})$ for large P , this means that roughly $d \geq 2P \ln N$ dimensions are needed to reliably code for Y .

Now, one very simple way to construct maps F and G is to choose the codewords $F(i)$ randomly from some distribution on the unit sphere in $\mathbb{R}^d$ and let $G(v)$ return the element j so that $\langle v, F(j) \rangle$ is maximized. For brevity, we refer to (F, G) . This strategy can be motivated by the remarkable fact that random codebooks of size $\Omega(e^d)$ of d -dimensional codewords can have bounded interference, in the sense of the following result.

Proposition 3 *Let (F, G) be constructed according to the random codebook strategy, let $\epsilon > 0$, and let*

$$d \geq (2 + \epsilon)P \ln N.$$

Then, when the random variable (Y, Z) are in the conditions of Proposition 2, $P(G(F(Y) + Z) = Y)$ converges to 1 for large N .

Altogether, this shows that transmitting

Our discussion so far parallels many Cover & Thomas (2006). In the remainder of this work, we consider

It will also be useful to know the upper bound

$$H = \ln \binom{N}{k} \leq k \ln(eN/k) = k \ln N - k \ln k + k$$

on the entropy of a random k -element subset of $[N]$, which turns out to be a very good approximation when $k \ll N$. For example, when $N = 2^{20}$ and $k = 2^8$, the approximation

$$\ln \binom{2^{20}}{2^8} \approx 2^8 \ln(2^{20}e/2^8) = 128(1 + 12 \ln 2)$$

holds with a relative error of only about 0.3%. (See ?? for a discussion of this estimate.)

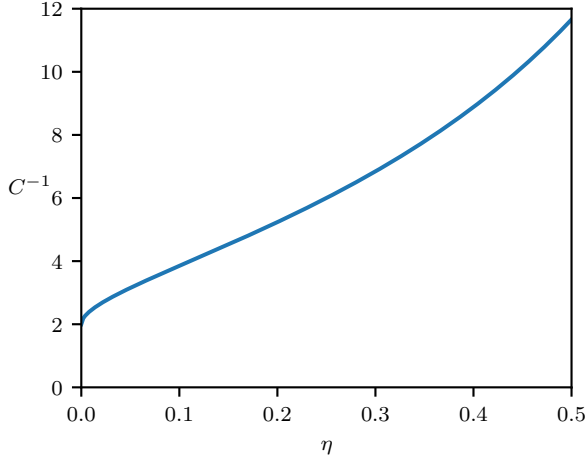


Figure 1: The upper bound on inverse channel capacity (in nats per dimension) for $\eta \in (0, 1/2)$ predicted by Proposition 5.

2.1 Results for Random Dictionaries

Our first result on random dictionaries provides an

Proposition 4 *Let F and G be as above, and let $C < 2$. Then for sufficiently large N , $P(G(F(Y)) = Y) < 1/2$.*

This shows that

Proposition 5 *For $\epsilon \in (0, 1)$, let $\ln k \leq \epsilon \ln N$, and let*

$$\kappa > 2 + 4\sqrt{\epsilon} + 2\epsilon.$$

Suppose $d \geq \kappa \ln N$. Then for sufficiently large N , $G(F(Y)) = Y$ with arbitrarily high probability.

Since

$$2 + 4\sqrt{\epsilon} + 2\epsilon \leq 4(1 + \epsilon)$$

and when $k = N^\epsilon$, $4(1 + \epsilon)k \ln N = 4k \ln(N/k)$, Proposition 5 shows that.

This prediction agrees well with numerical experiments, graphed in ???. In fact, even as N varies over several orders of magnitude, the slightly weaker condition $d \geq 8k \ln N$ characterizes the regime where the set Y can be decoded with reasonably high probability by threshold decoding.

Top- k decoding performs significantly better but admits a similar “rule of thumb”: for all values of N trialed, $d = 4k \ln kN$ is very close to the smallest dimension needed for top- k decoding to succeed with high probability. See ?? for an informal derivation of this bound.

There are several ways to interpret this conclusion. On one hand, it means that one-step estimates are asymptotically “inefficient” in terms of required bitrate when k is moderately large compared to N . More specifically, in a regime where N goes to infinity but $\ln k / \ln N$ converges to 1, we predict that one-step estimates require the ratio d/H to diverge to infinity.

In particular, one-step estimates are asymptotically inefficient when $k/N \geq \epsilon$ for some positive ϵ . Indeed, to have $d \geq 4k \ln N$ in this case we would need $d = \Omega(N \ln(N))$, while the entropy of Y is only $O(N)$. In contrast, a hallmark result of compressive sensing implies that, when $k/N \leq \epsilon$, the vector y can be recovered from its image Fy under a random projection by a certain *convex* optimization problem so long as $d \geq \kappa(\epsilon)N$ for some constant $\kappa(\epsilon)$; for example, see Candes & Tao (2005). The failure of our one-step estimates in this particular regime is easy to prove.

On the other hand, in a sparser regime where $\ln k / \ln N < \epsilon$ for some $\epsilon < 1$, it follows from our analysis that one-step estimates are “information-efficient” in the sense that they can be decoded from superposition codes that achieve bitrates H/d larger than some positive δ . However, it is also of interest to have *non-asymptotic* information on the required bitrate. From ?? we find that one-step estimates need at least $4 \ln 2 \approx 2.8$ bits per dimension even for small k . When $k = 2^8$ and $N = 2^{20}$ this number rises to about 4.1, and the experiments of ?? show that this factor is in fact slightly optimistic. If we use threshold decoding instead of top- k , we need 8 dimensions per bit! Can other inference algorithms do significantly better?

There is an extensive literature on theory of compressive sensing. Reeves et al. (2019) shows that, in our language, superposition codes with a random dictionary are essentially optimal in the information-theoretic sense when ideal maximum-likelihood inference is used as the decoder. A series of earlier works (Joseph & Barron (2012; 2014); Rush et al. (2017)) on superposition codes also showed that, under some special conditions on y , certain decoding schemes admit bitrates up to theoretical channel capacity in the presence of Gaussian noise. However, to our knowledge, practical guarantees on the performance of iterative methods are not available for our range of k and N .

?? shows the results of a numerical experiment using an iterative method called *matching pursuit*, first suggested in Bergeaud & Mallat (1995). This is a simple “greedy” algorithm that initializes $y = 0$ and, at each of k iterations, increments the index of y whose corresponding codeword has largest inner product with $x - Fy$.

Matching pursuit far outperforms top- k decoding for the range of N and k considered earlier. When $d \geq 1.3 \log_2(eN/k)$ and $N > 2^{16}$, our experiments show that matching pursuit is very reliable. In other words, matching pursuit can reliably infer a sparse vector from just 1.3 dimensions per bit. In ?? we find that a more computationally expensive algorithm called basis pursuit can do slightly better, requiring only 0.8 dimensions per bit.

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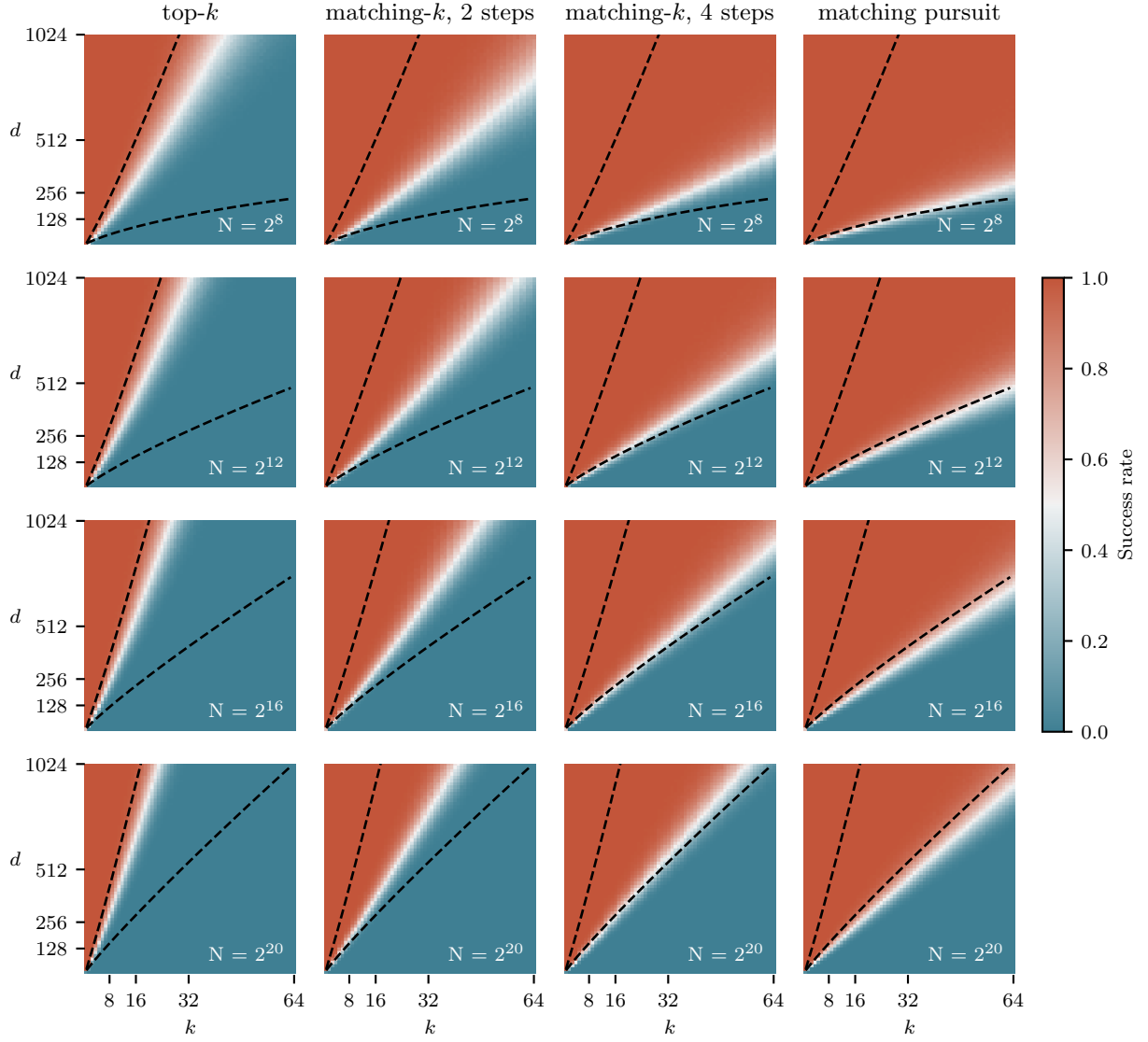


Figure 2: Empirical performance of a variety of different decoding methods at reading a k -element subset of $[N]$ from a d -dimensional superposition code with a Rademacher dictionary.

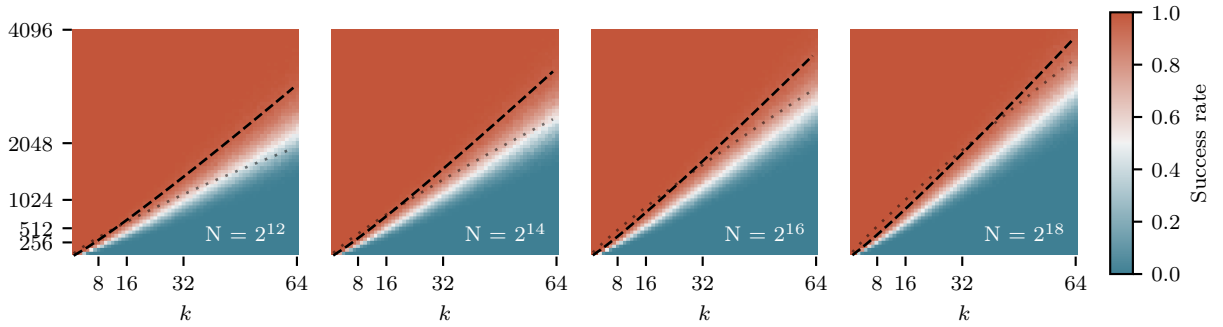


Figure 3: The prediction of Proposition 5 agrees well with numerical experiments.

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A Matched Filters

Consider the problem of inferring a scalar S from the sum

$$X = Sf + Z$$

where $f \in \mathbb{R}^n$ and Z is a Gaussian variable independent from S . Suppose for simplicity that Z has non-singular covariance Σ , so that $-\ln p(z) = 1/2\|z\|_\Sigma^2$ where

$$\|z\|_\Sigma^2 = z^T \Sigma^{-1} z.$$

Then a routine calculation shows that

$$\begin{aligned} & -\ln p(S = s | X = x) \\ &= C(x) - \ln p(s) + \frac{1}{2} \left(s - \frac{\langle f, x \rangle_\Sigma}{\|f\|_\Sigma^2} \right)^2 \|f\|_\Sigma^2 \end{aligned} \quad (3)$$

where $C(x)$ is a constant depending only on x and $\langle -, - \rangle_\Sigma$ is the inner product associated with the norm $\|-\|_\Sigma$. In particular, the distribution of S conditional on X is only a function of the inner product $\langle f, X \rangle_\Sigma$. The **matched filter** for S is the linear function

$$\lambda(X) = \frac{\langle f, X \rangle_\Sigma}{\|f\|_\Sigma^2},$$

and can be understood as providing the maximum likelihood estimate for S .

The quality of our matched filter is measured by its signal-to-noise ratio (SNR)

$$\rho = \frac{(\lambda(f))^2}{\text{Var}_Z \lambda(Z)} = \|f\|_\Sigma^2.$$

Up to a scalar, λ can be characterized as the linear function that maximizes this quantity. Given an improper uniform prior on S , Equation (3) shows the posterior on S conditional on X is a Gaussian with mean $\lambda(X)$ and precision ρ .